

Simon Hathout Willard

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Third year Engineering Physics student at UBC with skills in CAD, software development, and engineering design. Strong computer skills and able to pick up new software and tools quickly.

TECHNICAL EXPERIENCE

Automation Engineering Intern at Dometic Vancouver | January – April 2025

- Designed and implemented SCADA software using Ignition software, reducing scrap product and improving machine scheduling, leading to savings of \$15000 in 4 months
- Wrote Python (Jython) code to process data and create interactive Human Machine Interfaces through Ignition's Vision and Perspective modules.
- Used SQL (msSQLserver) to store machine data effectively, preserve ACID principles, and enforce relevant constraints on data.

Automation Subteam Member with Agropionics Design Team | 2023 - 2025

- Used IOT sensor systems with Arduino and ESP32 to measure PH and temperature in a nutrient solution, improving plant growth and productivity
- Wrote code to format data using JSON and transfer it wirelessly using MQTT
- Communicated with other subteams to determine required data, and assisted in selection and procurement of appropriate sensors
- Used Git and Github for sharing code and documentation, improving maintainability

NorthForge Design Space | July-August 2024

- Received formal training with woodshop and laser cutter tools, ensuring safe use.
- Learned the basics of web development and used these skills to create a portfolio, which can be found at <https://hathoutwillard.github.io/posts/>
- Worked part time as a contractor with Maker Mind Toys, sanding and ensuring consistent quality of wooden toys for children.

SKILLS

Programming - Java, SQL, C++, Python, HTML/CSS, Hugo SSG, Arduino

Design Tools - Fusion 360, Onshape, Machine Shop, 3D Printing, Laser Cutting, Woodshop

Software - Git and Github. Microsoft and Google suites, particularly Excel.

EDUCATION

3rd Year Engineering Physics | 2023 – Present | UBC Applied Sciences

- Courses include software development, engineering design, thermodynamics, and electric circuit design. A full list can be found at <https://hathoutwillard.github.io/courses/>
- Worked in a team of 4 to design and fabricate a robot with a three-axis arm, to autonomously navigate a course and retrieve objectives.
- Cumulative average: 93.2% (A+)